## OS4-1

# Individual variation in humpback whale signal production: the role of source-filter characteristics for individual identity encoding in a vocal learning mammal

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The variation in signal production that is needed for acoustic individual identification in mammals is often attributed to either source-filter or spectro-temporal pattern variation. The humpback whale is a baleen whale that is well known for its elaborate and diverse songs, which show evidence of vocal learning through gradual transitions of their seasonal songs which change year to year. Humpback whales also utilize social calls, which are stable across years and have been shown to be shared across populations on a global scale. This social sound repertoire is commonly used in social interactions among whales, such as during group feeding events on the foraging grounds. This study explores source-filter based variation in production of social calls by humpback whale to assess the potential for acoustic individual identification on the foraging grounds through use of a shared repertoire. Data were collected using acoustic biologging tags between 2007-2024 with 20 individual whales that were tagged in two or more years. We describe variation and stability of individual signatures in social call types and discuss the potential for a combination of source-filter and spectro-temporal pattern variation to convey individual specific information in this species.

## OS4-2

# Acoustic communication networks mediate recruitment for collective action in spotted hyenas

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Long-range acoustic signals may serve as critical tools for coordinating behavior in spatially dispersed animal societies, yet their network-level dynamics and functional roles are rarely quantified. Spotted hyenas live in large fission-fusion societies, where highly dispersed group-mates need to gather to cooperatively protect communal territories and resources within them. To investigate the dynamics of long-range acoustic recruitment in spotted hyena societies, we tracked a group of hyenas in the Maasai Mara using sound, movement and GPS-recording tags attached to 19 adult individuals. We used a custom CNN to detect long-range recruitment signals, termed whoops, and identified whoops from the tagged individual through throat vibrations detected by accelerometers. During the 3-week recording period, tagged animals produced 9338 whoops across some 1161 whoop bouts. We show that whoops can recruit conspecifics over distances of >3km. Notably, a subset of receivers respond not only by approaching but also by producing whoops themselves. We show that these secondary vocal responses serve multiple critical functions: they relay recruitment information across broader spatial scales to extend the collective recruitment range to a larger part of the communal territory; and they increase the information available within the communication network, leading to higher recruitment probability of 3rd party receivers. Our findings demonstrate how long-range acoustic signals and acoustic signaling cascades can enable rapid coordination at the scale of an entire group, highlighting the role of vocal networks for enabling collective action in distributed animal societies.

## OS4-3

# Kids are voice experts: early development of vocal modulation abilities

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People can modulate their voices to convey biologically relevant information (e.g., body size, age, and sex) during social interactions. While vocal modulation has been widely studied in adults, its developmental trajectory remains unclear. Research suggests that children can manipulate sexually dimorphic acoustic dimensions (e.g., fo) in speech to express gender-related attributes, but it is unknown whether they can extend these abilities to other physical traits, and how these abilities develop. To address this, we recorded 73 English-speaking children (5–11 yrs.) providing voices for animal characters varying in age, sex, and size, and then measured duration, fundamental frequency (fo), pitch modulation (fo CV), apparent vocal tract length (VTL), and roughness (HNR). Children systematically modulated their voices based on the characteristics of the animals. Both girls and boys imitated larger animals using longer duration, lower fo, and extended VTL, with boys elongating their vocal tract more than girls, and only girls reducing harmonicity. Children also adjusted vocal features to reflect the animals' sex, using shorter vocalizations, higher fo, greater fo variability, shorter VTL, and increased harmonicity for female animals, particularly in younger children and girls. When impersonating older animals, children spoke more slowly, with lower fo and reduced harmonicity. Most of these modulations align with actual vocal patterns observed in humans and other vertebrates. Our findings show that children develop early general vocal modulation abilities which they can easily deploy when imitating animal vocalizations.

## OS4-4

# Being social and vocal: social and vocal complexity in a cooperatively breeding passerine, Jungle Babbler

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Two important drivers proposed for the evolution of vocal complexity are sexual selection and sociality. While sexually-selected signals can be structurally complex, being display vocalizations, they are functionally limited. Social animals convey diverse information through their vocalizations. Consequently, they are likely to display both structural and functional complexity in their calls. At the same time, animals must minimize learning and production costs of signals, while keeping them sufficiently complex to retain functionality. Using bioacoustics and behaviour as our research tools, we examined vocal complexity and efficiency in the vocalization of a cooperatively breeding passerine, Jungle Babbler (Argya striata). We followed multiple groups of free-ranging Jungle Babblers and recorded and analysed calls after ascertaining the context of vocalizations to test the social complexity hypothesis. Recordings of 'Contact calls' made from identified individuals were analysed using a machine learning framework to test for evidence of sex and individual identity. We also examined their 'Alert calls' to test if the calls adhere to linguistic laws of information compression. Our results uphold the predictions of the social complexity hypothesis. We also found evidence for sex and individual-specific signatures in their 'Contact calls'. Lastly, we found evidence for efficiency in terms of information compression as dictated by linguistic laws in their 'Alert calls'. We argue that in our pursuit to understand the nuances of language evolution, one must examine both vocal complexity and efficiency in the contextual vocalizations of social animals, instead of display vocalizations of solitary species.

## OS4-5

# The clue is in the chorus

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Sound is an important cue to almost all marine fauna. Nearly two-thirds of all (≈35,000) fish species are predicted to produce sound and many of them do so with associated life functions (e.g., feeding, spawning, territorial, alert). Fish choruses are mass phenomena, created when many fish produce sound together, thereby increasing ambient noise levels for prolonged periods. The fish behaviour and, therefore, the resulting choruses, can be driven by environmental conditions, such as lunar phase, tidal cycle, time of sunset, temperature, salinity, and cholorphyll (among other variables), as well as extreme events, such as cyclones and anoxic events. We have documented over 300 fish choruses of over 80 different types in Australia's waters, in many cases speculating the associated behaviours. We will present a selection of these choruses to illustrate how long-term passive acoustic monitoring can tease out patterns in behaviour, abundance, and distribution of soniferous fishes, providing useful information for management and conservation, including their response (or lack of) to disturbance events.

## OS4-6

# Bird or cheetah? – Utilizing playbacks to test prey species reaction to cheetah chirps

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Cheetahs (Acinonyx jubatus) emit high-pitched vocalizations known as “chirps” that resemble bird calls in sound quality and structure when calling for conspecifics. The frequency of these chirps is considerably higher than expected for an animal of this size and for a sound employed in long-distance communication within a savannah habitat. Cheetahs are predators that rely on successful hunting attempts, while also facing threats from stronger predators like lions, which are among the main causes of mortality, particularly in cheetah cubs. The cheetah chirp is suggested to represent acoustic crypsis, mimicking avian calls and thereby hindering detection by prey species or competitors. To initially examine this idea, we performed playback studies with prey species, Impalas (Aepyceros melampus), Wildebeests (Connochaetes taurinus), Zebras (Equus quagga) and Kudus (Tragelaphus strepsiceros), at the Mabula Game Reserve in South Africa in September and October 2024. We deployed cheetah chirps, various other cheetah vocalizations, calls from other predators, and bird calls as controls, broadcasting these stimuli via loudspeaker from one vehicle, while simultaneously video recording the reactions of the prey species from a second vehicle. Videos were coded for orientation and movement response as well as latency to look and move. I will be presenting the current state of the project and discuss some first preliminary results.

## OS4-7

# First evidence for widespread sharing of stereotyped non-signature whistle types by wild dolphins

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Over the past 41 years, we have built a unique library of sounds produced by known individual common bottlenose dolphins (Tursiops truncatus), by recording them with suction cup hydrophones during brief catch-and-release health assessments and with digital acoustic tags (DTAGs). We have catalogued the individually distinctive signature whistles (SWs) of most animals in this resident community of 170 dolphins, which has enabled us to begin studying poorly defined "non-signature whistles" (NSW). We have so far identified 22 shared NSW types, two of which (NSWA and NSWB) are known to have been produced by at least 25 and 35 different dolphins respectively. Several of these NSWs contain features that are rarely seen in SWs, including constant frequency components and short, steep, up and down sweeps. We are studying the functions of shared NSWs through playback experiments to free-swimming dolphins. Varied responses to playbacks reflect the complexity of dolphin communication, and highlight the need for larger sample sizes to correctly interpret NSW functions. However, results so far have provided support for the affiliative nature of SW copies, with a majority of control playbacks of a dolphin's own signature whistle eliciting positive responses (turn toward the speaker). NSWA elicited a majority of negative responses (turn away from the speaker), suggesting an alarm-type function. NSWB elicited varying responses; we hypothesize that this whistle may be produced when something unexpected or unfamiliar is heard. Our study provides the first evidence for widespread sharing of stereotyped and potentially context-specific non-signature whistles in a dolphin community.

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## OS4-8

# Calling out loud: influence of land-use change, social context, climate, and body condition on amphibian calling behaviour in the Western Ghats, India

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Amphibian calling behaviour is shaped by a complex interplay of environmental, physiological, and social factors. Among anthropogenic pressures, land-use change is a major driver of biodiversity loss, often altering morphological and behavioural traits that affect species fitness, yet remain poorly understood. While temperature and rainfall are known to influence calling behaviour, local calling density can also play a role through vocal competition. To disentangle these effects, we examine the relative influence of neighbour calling density, rainfall, air temperature, and body condition on the calling behaviour of arboreal frogs, and how these relationships vary across land-use types. We recorded advertisement calls of four arboreal frog species from June to September 2024 in relatively undisturbed rainforests and coffee agroforests in the Western Ghats. Our dataset includes 10-minute focal recordings from 250 individuals, accompanied by measurements of air temperature, humidity, rainfall, body size, body mass, and the number of calling conspecific and heterospecific males within a 15-meter radius. Using automated note-level annotation, we extracted temporal (call rate, duration, inter-call interval) and spectral (dominant frequency) properties of calls. Preliminary results indicate that air temperature and humidity did not differ significantly across land-use types. Surprisingly, all four species exhibited better body condition in coffee agroforests than in rainforests. We are currently applying structural equation modelling (SEM) to evaluate the interacting effects of intrinsic and extrinsic drivers on calling behaviour. The findings will help us better understand the behavioural consequences of land-use change and the sensitivity of endemic amphibians to future biotic and abiotic changes.

## OS4-9

# Social functions of vocalization in southern white rhinoceros

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Animal vocalizations receive more and more interest in animal management and conservation since vocalizations encode a variety of information about the sender and can alter behavioral responses in recipients. Therefore, we investigated the vocal repertoire and behavioural reactions towards conspecific vocalizations in the Southern white rhinoceros, an endangered species, suggested to have the most complex vocal repertoire among rhinoceros' species. Audio and video data from 11 European zoos and two field sites in Africa (Botswana and Uganda) were recorded and an acoustic and behavioural analysis was performed. Additionally, playback experiments were conducted to investigate how the rhinoceros respond to conspecific vocalizations. Comparing captive and field data, we found the same call types supporting that findings in captivity reflect the animals' natural vocal repertoire. Focussing on the most common call types, vocal exchanges reflect the social relationships among herd members and showed sex-specific pattern for both captive and free-ranging animals. Playback experiments revealed that rhinoceros discriminate between conspecific call types. Thereby, rhinoceros showed the strongest responses towards conspecific Pant calls, which are suggested to function as contact and mating call. Overall, the results highlight that rhinoceros vocalizations encode information about the social and reproductive dynamics and elicit specific behavioral responses, making them to a valuable tool for conservation and management strategies.

## OS4-10

# Characterizing song-like vocal behavior in the naked mole-rat: individuality, syntax, and context

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Naked mole-rats, a highly social rodent species, rely heavily on vocal communication across a wide range of social contexts. During courtship, vocal exchanges occur between the queen, the sole breeding female, and breeding males. To investigate the function and structure of vocalization in mate selection and pair bonding, we conducted continuous behavioral monitoring of naked mole-rats as they transitioned to active breeding. To establish new naked mole-rat colonies and eliminate reproductive suppression, we isolated individuals of both sexes and housed them separately for two months with weekly pairings for brief periods of increasing duration. We made continuous audio recordings of isolated individuals and additionally made video recordings of weekly pairings. Furthermore, we investigated both individual- and colony-specific variation in male vocalizations. Our analysis revealed consistent individual differences in ordering of vocal elements, as well as context-dependent variation in vocal behavior. These findings suggest that male vocal production is shaped by both intrinsic individual traits and extrinsic social or environmental factors.